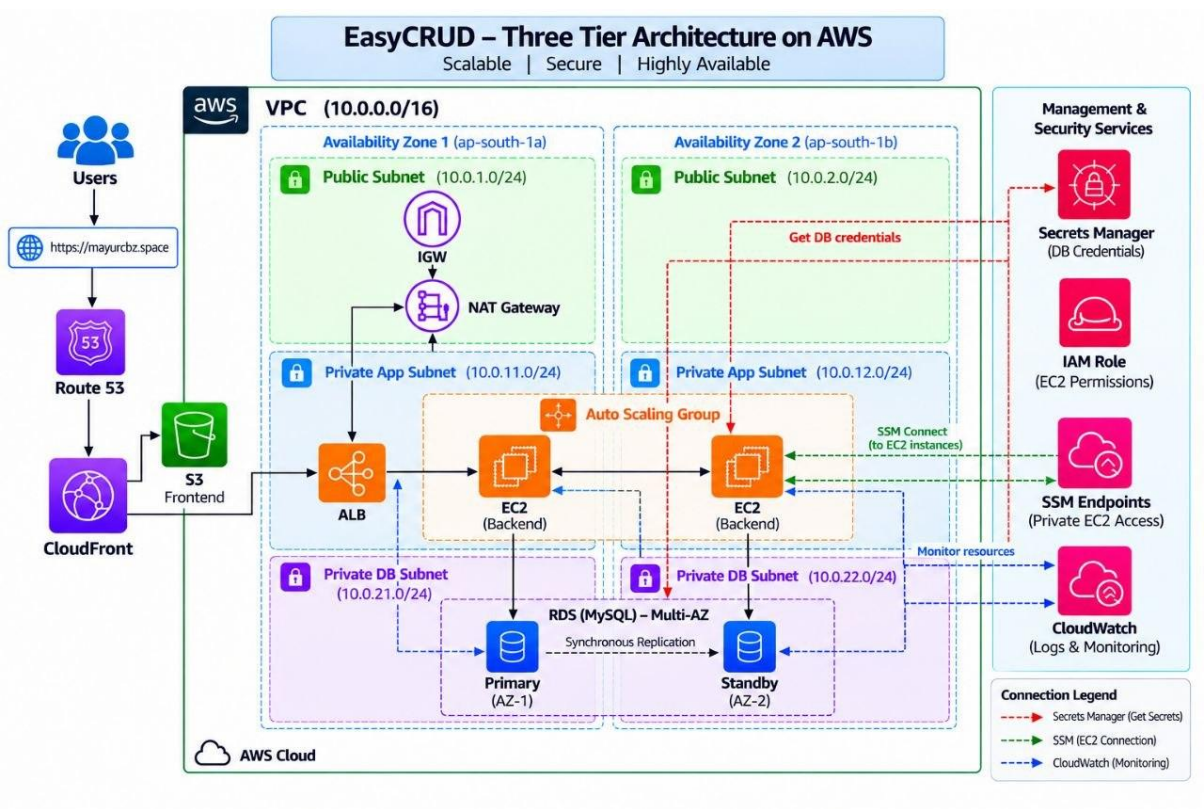
EasyCRUD – THREE TIER ARCHITECTURE ON AWS



We separate the infrastructure into three logical layers:

1. Public subnet → internet-facing resources
2. Private application subnet → backend servers
3. Private database subnet → database

PHASE-1 VPC

1.create a vpc

1. name -- easycrud-vpc
2. cidr ---- 10.0.0.0/16

2.create a subnet

1. public-subnets
    1. easycrud-public-1 --- 10.0.1.0/24
    2. easycrud-public-2 --- 10.0.2.0/24
2. private-subnets
    1. easycrud-private-app-1 --- 10.0.11.0/24
    2. easycrud-private-app-2 --- 10.0.12.0/24
3. private-database-subnets
    1. easycrud-private-db-1 --- 10.0.21.0/24
    2. easycrud-private-db-2 --- 10.0.22.0/24

3.Enable public IPv4 - public-subnet-1

1. click on public-subnet-1
2. go to actions
3. click on edit subnet settings
4. enable -- auto-assign public IPv4 address
5. save

4.Enable public IPv4 - public-subnet-2

1. click on public-subnet-2
2. go to actions
3. click on edit subnet settings
4. enable -- auto-assign public IPv4 address
5. save

5.create an internet gateway (easycrud-igw) and attach to vpc

6.create a public-route-table

1. name --- easycrud-public-rt
2. route
    1. destination -- 0.0.0.0/0
    2. target --------- easycrud-igw
3. associate public subnets
    1. easycrud-public-1
    2. easycrud-public-2

5.create a nat gateway

1. name ----- easycrud-nat
2. subnet --- easycrud-public-1
3. connectivity -- public
4. allocate elastic ip address
5. create a nat gateway

6.create a private route table

1. name --- easycrud-private-app-rt
2. route
    1. destination -- 0.0.0.0/0
    2. target --------- easycrud-nat
3. associate private subnets
    1. easycrud-private-app-1
    2. easycrud-private-app-2

7.create a database route table

1. name ----- easycrud-private-db-rt
2. associate
    1. easycrud-private-db-1
    2. easycrud-private-db-2

PHASE – 2

8.create a role

1. name ---- easycrud-role
2. entity -- EC2
3. permissions attach
    1. AmazonSSMManagedInstanceCore
    2. CloudWatchAgentServerPolicy
4. create a role

9.Add secrets manager permission

1. create a inlined policy in easycrud-role
2. add the json policy

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ReadEasyCRUDRDSSecret",
      "Effect": "Allow",
      "Action": [
        "secretsmanager:GetSecretValue"
      ],
      "Resource": "YOUR_RDS_SECRET_ARN"
    }
  ]
}
```

{Replace: YOUR_RDS_SECRET_ARN}

3. policy name - easycrud-read-rds-secret
4. create policy

9.create a SSM vpc Endpoints

Because the backend instances are private, we configured VPC endpoints so that SSM can work without a public IP.

1. Enable vpc

10.Enable VPC DNS Support

1. click on easycrud vpc
2. go to actions -- edit vpc settings
3. DNS resolution -- enable
4. DNS hostnames -- enable
5. save changes

11.Create VPC Endpoint Security Group

1. go to security group
2. create a security group

3. name ---- easycrud-vpce-sg
4. Description → EasyCRUD VPC Endpoint SG
5. vpc ----- easycrud-vpc
6. Inbound Rules
    1. type ---- https
    2. port ---- 443
    3. source -- custom
    4. CIDR ---- 10.0.0.0/16
7. Outbound Rule
    1. type ---- all traffic
    2. Destination --- 0.0.0.0/0
8. create security group

12.Create SSM Endpoint

1. Go to endpoint
2. create a endpoint
3. Name → easycrud-ssm-endpoint
4. Service category → AWS services
5. Service → com.amazonaws.ap-south-1.ssm
6. VPC → easycrud-vpc
7. Endpoint type → Interface
8. subnets
    1. easycrud-private-app-1
    2. easycrud-private-app-2
9. security group ---- easycrud-vpce-sg
10. policy --- full access
11. create a endpoint

13.Create EC2 Messages Endpoint

1. Go to endpoint
2. create a endpoint
3. Name → easycrud-ec2messages-endpoint
4. Service category → AWS services
5. Service → com.amazonaws.ap-south-1.ec2messages
6. VPC → easycrud-vpc
7. Endpoint type → Interface
8. subnets
    1. easycrud-private-app-1
    2. easycrud-private-app-2
9. security group ---- easycrud-vpce-sg
10. policy --- full access
11. create a endpoint

14.Create SSM Messages Endpoint

1. Go to endpoint
2. create a endpoint
3. Name → easycrud-ssmmessages-endpoint
4. Service category → AWS services

5. Service → com.amazonaws.ap-south-1.ssmmessages
6. VPC → easycrud-vpc
7. Endpoint type → Interface
8. subnets
    1. easycrud-private-app-1
    2. easycrud-private-app-2
9. security group ---- easycrud-vpce-sg
10. policy --- full access
11. create a endpoint

15.Create S3 Gateway endpoint

1. Go to endpoint
2. create a endpoint
3. Name → easycrud-s3-endpoint
4. Service category → AWS services
5. Service → com.amazonaws.ap-south-1.s3
6. VPC → easycrud-vpc
7. Endpoint type → Gateway
8. Route tables ------ easycrud-private-app-rt
9. policy --- full access
10. create a endpoint

PHASE – 3 RDS & SECRETS MANAGER

1.Create a DB subnet group

1. name ---- easycrud-db-subnet-group
2. Description → EasyCRUD DB subnet group
3. VPC → easycrud-vpc
4. Availability Zones →
    1. ap-south-1a
    2. ap-south-1b
5. Subnets →
    1. easycrud-private-db-1
    2. easycrud-private-db-2
6. create a subnet group

2.create a RDS Database

1. machine name ---- MySQL
2. DB instance identifier → easycrud-rds
3. username ----- admin
4. credentials ---- Managed in AWS Secrets Manager
5. Compute resource → Don't connect to an EC2 compute resource
6. VPC → easycrud-vpc
7. DB subnet group → easycrud-db-subnet-group
8. Public access → No
9. VPC security group → easycrud-rds-sg

10. Initial database name → student_db
11. Backup retention period → 7 days
12. create a database

3.Get the Secrets Manager Secret

1. go to secrets manager
2. click on the secrets
3. You get
    1. username
    2. password
    3. engine
    4. host
    5. port
    6. dbname
4. Important
    1. We do not copy the password into:
        - EC2 user data
        - application.properties
        - GitHub
        - source code

4.Go to Security Group

1. click on easycrud-rds-sg
2. edit inbound
    1. type --- MySQL/Aurora
    2. source - easycrud-backend-sg

At this point, if easycrud-backend-sg hasn't been created yet, don't create a 0.0.0.0/0 rule.
We'll create easycrud-backend-sg in the backend phase and then add it as the source.

PHASE - 4 BACKEND CONFIGURATION – SSM

1.create a backend-security-group

1. sg name → easycrud-backend-sg
2. Description → EasyCRUD Backend Security Group
3. VPC → easycrud-vpc
4. inbound rules
    1. type -- custom TCP
    2. port -- 8080
    3. source - custom
    4. source -- 10.0.0.0/16
5. create a sg

Later, after creating the ALB security group, we will change this rule to allow 8080 only from easycrud-alb-sg.

2.Update RDS security group

1. click on easycrud-rds-sg & edit these rule

2. type ---- MySQL/Aurora
3. port ---- 3306
4. source -- custom
5. soure --- easycrud-backend-sg
6. create a sg

3.Create a Backend EC2

1. name ---- easycrud-backend-1
2. os -------- amazon-Linux 2023
3. type ----- t3.micro
4. key pair - Proceed without a key pair
5. vpc ------ easycrud-vpc
6. subnet --- easycrud-private-app-1
7. public ip - disable
8. sg name --- easycrud-backend-sg
9. Iam instance profile -- easycrud-role
10. launch instance

4.Connect using SSM

1. Go to SSM
2. click on easycrud-backend-1
3. connect to terminal

5.Configuration on easycrud-backend-1

1. dnf update -y
2. sudo dnf install java-17-amazon-corretto -y
3. sudo dnf install git -y
4. sudo dnf install maven -y
5. java --version
6. git --version
7. aws --version
8. mvn --version
9. cd /opt
10. git clone https://github.com/mayurpandit25/EasyCRUD-Updated-df.git
11. sudo chown -R ssm-user:ssm-user /opt/EasyCRUD-Updated-df
12. sudo find /opt/EasyCRUD-Updated-df -type d -exec chmod 755 {} \;
13. sudo find /opt/EasyCRUD-Updated-df -type f -exec chmod 644 {} \;
14. cd /opt/EasyCRUD-Updated-df/
15. vim backend/src/main/resources/application.properties

```
server.port=8080
spring.datasource.url=jdbc:mariadb://${DB_HOST}:${DB_PORT:3306}/${DB_NAME}?sslMode=trust
spring.datasource.username=${DB_USER}
spring.datasource.password=${DB_PASSWORD}
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=false
management.endpoints.web.exposure.include=health,info
management.endpoint.health.show-details=never
```

16. cd /opt/EasyCRUD-Updated-df/backend
17. mvn clean package -DskipTests
18. ls target/

6.Tests Secrets Manager Access

1. sudo aws secretsmanager get-secret-value --secret-id "YOUR_RDS_SECRET_ARN" --region ap-south-1

7.Set Database Environment Variables Commands

1. export DB_HOST="YOUR_RDS_ENDPOINT"
2. export DB_PORT="3306"
3. export DB_NAME="student_db"
4. export DB_USER="admin"

Replace: (Your_RDS_Endpoint)

8.Get the password from Secrets Manager:

```
export DB_PASSWORD=$(aws secretsmanager get-secret-value \
 --secret-id "YOUR_RDS_SECRET_ARN" \
 --region ap-south-1 \
 --query SecretString \
 --output text | python3 -c 'import sys,json; print(json.load(sys.stdin)["password"])')
```

9.Start Backend

1. cd /opt/EasyCRUD-Updated-df/backend
2. nohup java -jar target/student-registration-backend-0.0.1-SNAPSHOT.jar > backend.log 2>&1 &
3. lsof -i :8080
4. tail -f backend.log

10.Test Backend Locally

1. curl http://localhost:8080/api/users

PHASE – 5 CREATION OF AMI

1.create a AMI From backend EC2

1. select instance --- easycrud-backend-1
2. click on Actions
3. click on Image and Templates
    1. Image name → easycrud-backend-ami
    2. Image description → EasyCRUD Backend AMI
4. create image

PHASE – 6 AUTOSCALING

1.Create launch template

1. Launch template name → easycrud-backend-lt
2. Template version description → EasyCRUD backend
3. AMI ---- My AMIs
4. AMI ---- easycrud-backend-ami

5. type --- t3.micro
6. Key pair → Don't include in launch template
7. subnet → Don't include in launch template
8. Auto-assign public IP → Disable
9. sg name --- easycrud-backend-sg
10. IAM instance profile → easycrud-role
11. user-data

```
#!/bin/bash

SECRET_ARN="YOUR_RDS_SECRET_ARN"

DB_HOST="YOUR_RDS_ENDPOINT"
DB_PORT="3306"
DB_NAME="student_db"
DB_USER="admin"

DB_PASSWORD=$(aws secretsmanager get-secret-value \
  --secret-id "$SECRET_ARN" \
  --region ap-south-1 \
  --query SecretString \
  --output text | python3 -c 'import sys,json; print(json.load(sys.stdin)["password"])')

cat > /etc/easycrud-backend.env <<EOF
DB_HOST=$DB_HOST
DB_PORT=$DB_PORT
DB_NAME=$DB_NAME
DB_USER=$DB_USER
DB_PASSWORD=$DB_PASSWORD
EOF

chmod 600 /etc/easycrud-backend.env

cat > /etc/systemd/system/easycrud-backend.service <<'SERVICE'
[Unit]
Description=EasyCRUD Spring Boot Backend
After=network-online.target
Wants=network-online.target

[Service]
User=ec2-user
WorkingDirectory=/opt/EasyCRUD-Updated-df/backend
EnvironmentFile=/etc/easycrud-backend.env
ExecStart=/usr/bin/java -jar /opt/EasyCRUD-Updated-df/backend/target/student-registration-backend-0.0.1-SNAPSHOT.jar
Restart=always
RestartSec=10

[Install]
WantedBy=multi-user.target
SERVICE

systemctl daemon-reload
systemctl enable easycrud-backend
systemctl start easycrud-backend
```

12. Replace

1. YOUR_RDS_SECRET_ARN
2. YOUR_RDS_ENDPOINT

13.Create a launch template

3.Create A auto scaling group

1. asg name --- easycrud-backend-asg
2. launch template --- easycrud-backend-lt
3. version ---- latest
4. vpc -------- easycrud-vpc
5. subnets
    1. easycrud-private-app-1
    2. easycrud-private-app-2
6. desired capacity -- 2
7. minimum capacity -- 2
8. maximum capacity -- 4
9. create a auto scaling group

PHASE – 7 LOAD BALANCING

1.create a ALB security group

1. Security group name → easycrud-alb-sg
2. Description → EasyCRUD ALB Security Group
3. VPC → easycrud-vpc
4. Inbound Rules
    1. type --- http
    2. port --- 80
    3. source - Anywhere-IPv4
    4. CIDR --- 0.0.0.0/0

2.Update Backend Security Group

1. edit the easycrud-backend-sg
2. type ---- custom TCP
3. port ---- 8080
4. source -- custom
5. source -- easycrud-alb-sg
6. save rules

3.Create a Target Group

1. type ------- instances
2. tg name ---- easycrud-backend-tg
3. protocol --- HTTP
4. Port ------- 8080
5. vpc -------- easycrud-vpc
6. health check protocol --- HTTP
7. Health Check path ------- /api/users
8. port ---- traffic port
9. create a target group

4.Add target group in the autoscaling group

1. go to autoscaling group
2. add the target group in the autoscaling group

5.Create a Application load balancer

1. lb name ---- easycrud-alb
2. scheme ----- internet-facing
3. vpc -------- easycrud-vpc
4. subnets
    1. easycrud-public-1
    2. easycrud-public-2
5. sg --------- easycrud-alb-sg
6. protocol --- HTTP
7. port ------- 80
8. forward to - easycrud-backend-tg
9. create a load balancer

6.Check the health of instance in target group

7.Test ALB

1. http://YOUR-ALB-DNS

PHASE - 8 CLOUDWATCH WITH SNS & AUTOSCALING POLICY

1.go to SNS

1.create a topic first (easycrud-app)

2.create a subscription

1. endpoint -- email
2. email end -- mayur@gmail.com

2.Go to CloudWatch

1. create a alarm

3.create a autoscaling policy

1. go to easycrud-backend-asg
2. click on asg
3. click on instance scaling
4. create a dynamic scaling policy
    1. name --- increase scaling policy
    2. alarm -- add the cloudwatch alarm
    3. add ---- 2
5. create a autoscaling policy

PHASE - 9 BUILD FRONTEND

1.Create a s3 bucket

mayurcbz.space

Build Frontend Locally

1. git clone https://github.com/mayurpandit25/EasyCRUD-Updated-df.git
2. cd EasyCRUD-Updated-df
3. cd frontend
4. vim .env
   VITE_API_URL=/api
5. npm install
6. npm run build
7. ls dist/
8. aws s3 cp dist/ s3://mayurcbz.space/
9. aws s3 ls s3://mayurcbz.space/
10. Enable static website hosting and test the frontend

PHASE – 10 {ACM - ROUTE-53 – CLOUDFRONT}

1.create a route 53 hosted zone

1. mayurcbz.space

2.Create a ACM certificates in North Virginia

1. domain name → mayurcbz.space
2. request certificates
3. After that create a record in route 53

3.Create CloudFront Distribution

1. Distribution name → mayurcbz.space
2. Description → mayurcbz.space
3. Route 53 managed domain → mayurcbz.space
4. origin type → amazon s3
5. s3 origin → select bucket (do not select the static website URL)
6. select the acm certificates arn
7. create a CloudFront distribution

4.Edit the CloudFront

1. Default root object ---> index.html

5.Edit the CloudFront

1. Alternate domain names ---> mayurcbz.space

6.Edit the CloudFront

1. go to origins
2. create a origin
   1. Origin domain ----> easycrud-alb (select a load balancer)
   2. origin name -------> easycrud-alb-origin
   3. create a origin

7.Edit the CloudFront

1. click on behaviour
2. create a behaviour
3. Path pattern → /api/*
4. Origin → easycrud-alb-origin
5. Viewer protocol policy → Redirect HTTP to HTTPS
6. Cache policy → CachingDisabled
7. Origin request policy → AllViewerExceptHostHeader
8. create a behaviour

8.Create a record in route 53

1. create a record
2. record type -- A
    1) alias
    2) select CloudFront Distribution
    3) region --- ap-south-1
    4) endpoint – (CloudFront-dns-name)
3. create a record

9. Test the Domain name

1. https://mayurcbz.space ---------> Frontend
2. https://mayurcbz.space/api/ ---> Backend
